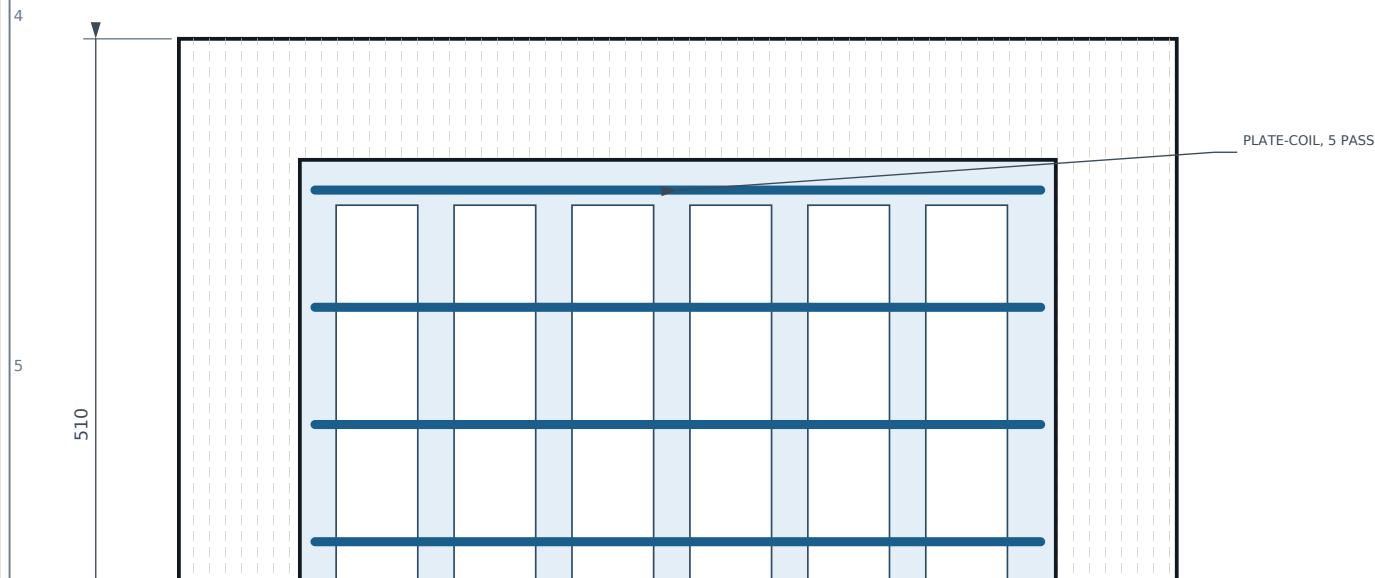
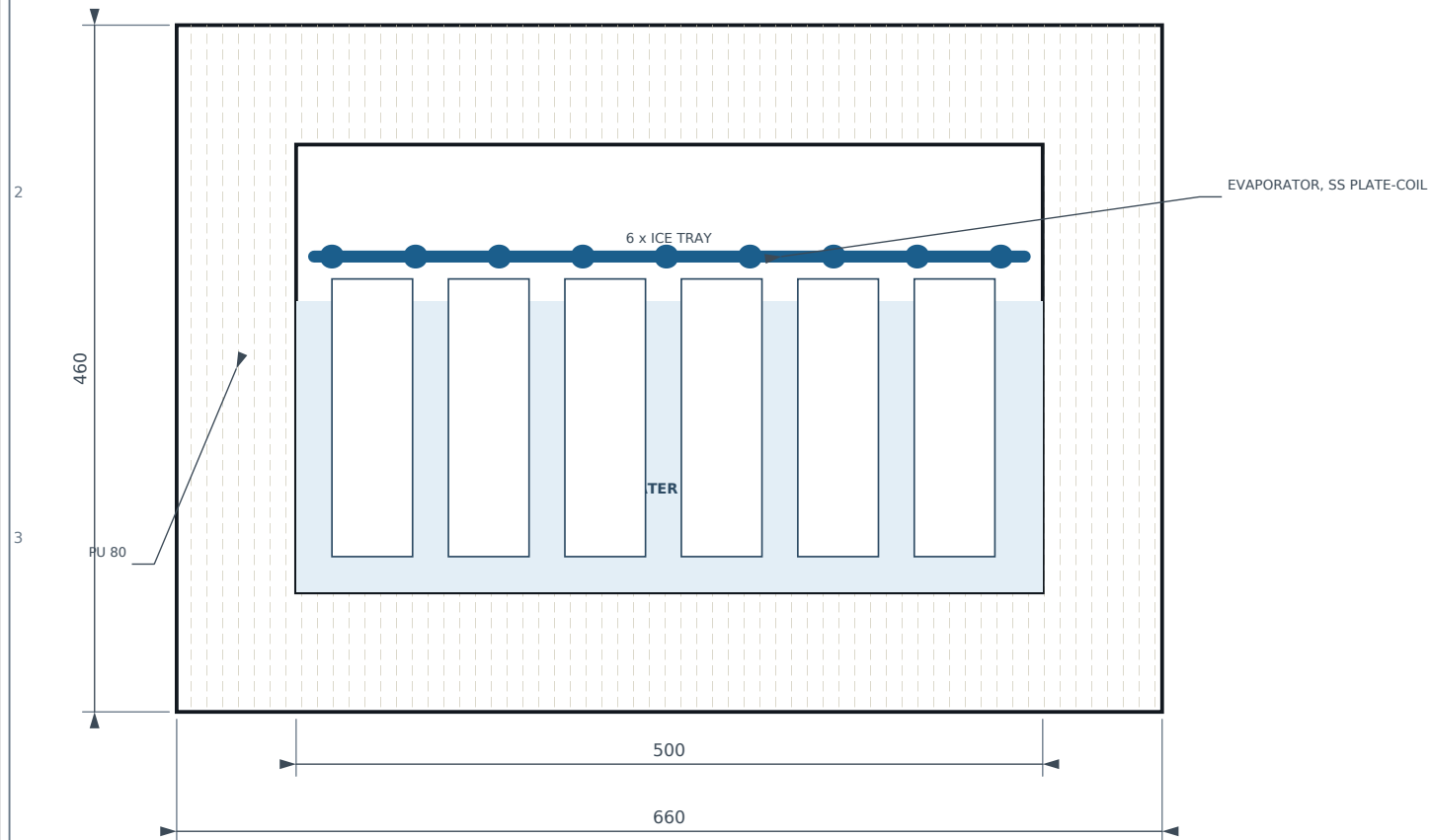


ICE BOX – SECTION

SCALE 1:5



INSPECTION		
CHECK	REQUIREMENT	METHOD
Evaporator external collapse	full vacuum, no set	1 bar external, 1 h
Helium leak	< 1e-8 Pa.m3/s	mass spectrometer
Containment tray	welded, drains to bund	water fill test
Box heat leak	<= 10 W at 30 C	calibrated pull-down

EVAPORATOR AND BOX DATA	
evaporating temperature	-5 C
evaporating pressure	2.92 kPa abs
cooling delivered per night	2.19 MJ
mean power over 12 h	51 W
surface required	508 cm ²
box internal	500 x 350 x 300
insulation U	0.30 W/m ² K
heat leak at 30 C ambient	8 W
leak as share of the night	15 %

NOTES

1. The evaporator runs at 2.9 kPa absolute. It sees full external atmospheric pressure whenever the machine is cold - design it as a vacuum vessel, not a pressure vessel.
2. Plate-coil in SS 316L, all welded, no fittings inside the box.
3. Evaporator sits ABOVE the trays so liquid methanol drains back to the receiver by gravity when the machine warms.
4. Secondary containment: the box is lined with a welded SS tray connected to the bund on SI-300. A leak must not reach the ice.
5. The ice is NOT potable. Methanol is on the other side of one wall. Use the ice for cooling, not for drinks.
6. 80 mm PU gives a 15 % overnight leak, already inside the 5 kg figure.

INDIA-VC THERMAL HARDWARE					ISO 128 / ISO 2768-mK	
TITLE EVAPORATOR AND ICE BOX 5 kg ICE PER NIGHT / 51 W MEAN / 80 mm PU			SCALE 1:5 THIRD ANGLE PROJECTION		UNITS mm	
DRAWING No. SI-104			REV A		SHEET 4 / 6	
DRAWN AI-ASSISTED DRAFT	CHECKED — PENDING —	MATERIAL SEE BOM	FINISH SEE NOTES		DATE 2026-09-15	